Project Proposal: Gaming Habits and Academic Performance

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Dataset

The dataset contains records for 8,000 students, with variables covering gaming behavior (daily gaming hours, genre, addiction score, device usage), lifestyle factors (study hours, sleep hours, attendance, social activity), physiological/psychological measures (reaction time, stress level), and academic outcome (grades).

Dataset link

https://www.kaggle.com/datasets/nalisha/gaming-vs-academic-performance

Problem Statement

Video games are often blamed for poor academic performance, but the relationship is likely more nuanced than "more gaming = worse grades." This project investigates which factors most strongly predict student grades. Specifically whether there are identifiable thresholds (cutoffs) in variables like gaming hours, study hours, and sleep hours beyond which grade outcomes shift noticeably (e.g., the point at which average grade drops from A- to B+).

Objectives

1. Identify which features (gaming hours, study hours, sleep hours, attendance, etc.) have the strongest association with grades
2. Determine threshold values within each feature where performance category changes (e.g., "under 4 hours of gaming/day is associated with A-range grades; above that, performance drops")
3. Explore whether these effects are independent or interact with each other (e.g., does high study time offset high gaming time?)

Dimensional Reduction Hypotheses

- Gaming-related variables (gaming hours, device usage, reaction time, addiction score) are expected to load heavily onto a shared component, since they likely reflect a single underlying "screen engagement" factor.
- Study-related and wellbeing variables (study hours, sleep, attendance) are expected to form a separate component distinct from the gaming cluster.
- Using CCA to relate a gaming/behavior variable set against a lifestyle/academic variable set, a moderate canonical correlation is expected, reflecting that gaming and lifestyle habits are related but not fully overlapping.

Clustering Hypotheses

- Unsupervised clustering is expected to reveal distinct student profiles (e.g., a high-gaming/low-study cluster vs. a low-gaming/high-study cluster) without being told grades in advance.
- Clusters formed on PCA-reduced features are expected to be more distinct/interpretable than clusters formed on the raw variables, since PCA should strip out noise and correlated redundancy first.

Methodology (planned)

1. **Data cleaning/prep —** check for missing values, outliers, and bin grades into letter-grade categories for threshold analysis.

2. **Exploratory analysis —** correlation matrix and scatterplots to see which variables move with grades.

3. **Threshold/cutoff detection —** approaches like decision trees (which naturally split continuous variables at breakpoints) or binned averages/boxplots per range to find where grade averages shift.

4. **Statistical testing —** regression (to quantify effect size and control for other variables at once) and/or ANOVA across binned groups to confirm differences are statistically significant, not just visual patterns.

5. **Interpretation —** summarize the "cutoff points" per feature in plain language.